

VeriCheck: A Real-Time Fact-Checking Browser Plugin for Vietnamese

Trinh Quoc Bao^{1,2}, Nguyen Khanh Tuan Anh^{1,2}, Pham Dang Khoa^{1,2}, Kiet Van Nguyen^{1,2*}

¹Faculty of Information Science and Engineering, University of Information Technology, Ho Chi Minh City, Vietnam

²Vietnam National University, Ho Chi Minh City, Vietnam

{22520055, 22520125, 22520685}@gm.uit.edu.vn, kietnv@uit.edu.vn

Abstract—We present VeriCheck, a real-time browser extension for fact-checking Vietnamese-language claims on social media. Designed to support users in identifying misinformation, VeriCheck processes user-provided claims through a multi-stage pipeline that combines advanced natural language processing techniques. The system first performs optimized query generation to retrieve relevant documents from credible Vietnamese sources, leveraging search engine APIs and applying source filtering heuristics. Retrieved texts are then reranked using a hybrid approach that integrates both sparse (BM25) and dense (SBERT) retrieval techniques. Finally, claim verification is conducted using transformer-based natural language inference models trained on the ViFactCheck dataset. Large language models are also incorporated for auxiliary tasks such as summarizing evidence and suggesting revised versions of misleading claims. A prototype implementation and user feedback demonstrate the practical potential of integrating automated fact-checking into user workflows, particularly for low-resource languages such as Vietnamese.

I. INTRODUCTION

The Vietnamese information sphere has undergone a rapid digital transformation: news is now consumed primarily through social platforms and messaging apps rather than traditional outlets. In such an environment, rumours and misleading claims can amass thousands of interactions before professional journalists or domain experts even notice their presence. This dynamic amplifies the impact of *everyday* factual errors—typos, misremembered dates, or incomplete context—that would once have been corrected in an editorial desk but now circulate unchecked across group chats and comment threads.

Manual verification remains the gold standard, yet it is infeasible at social-media scale. A single editor might validate only a handful of claims per hour, while an automated bot can spread hundreds of dubious posts in the same time. Consequently, recent research has turned to NLP-based fact-checking pipelines that combine evidence retrieval with natural-language inference; however, most publicly available systems and datasets target English, leaving Vietnamese users to rely on ad-hoc searches or proprietary tools.

Although the ViFactCheck benchmark has provided an essential training and evaluation resource, three gaps persist in practice: (i) a lack of high-recall evidence retrieval *from the live web* rather than fixed corpora, (ii) limited understanding of the latency–accuracy trade-offs among existing Vietnamese

veracity models, and (iii) minimal integration into the user’s natural reading or writing workflow.

We present *VeriCheck*, a lightweight, browser-integrated framework designed to address these gaps. When the user highlights a sentence, VeriCheck dispatches carefully reformulated queries to the **Serper Search API**—a JSON interface to Google Search—then re-ranks returned snippets with a Vietnamese cross-encoder and produces a SUPPORT/REFUTE verdict accompanied by token-level uncertainty. The entire round-trip completes in well under a second on commodity hardware, making the tool suitable for day-to-day use.

Our contributions are:

- 1) *Real-time retrieval*. VeriCheck relies on Serper Search to obtain fresh web evidence on demand, eliminating the maintenance cost of a local document index while still achieving competitive recall through a Vietnamese cross-encoder re-ranker.
- 2) *Comprehensive evaluation*. We benchmark a spectrum of Vietnamese veracity classifiers—from a 110 M-parameter RoBERTa to 7 B-parameter instruction-tuned LLMs—exposing the latency / accuracy trade-offs that guide deployment choices on ViFactCheck.
- 3) *Gemini-powered corrections*. A Gemini LLM module generates succinct rationales and proposes revised wording for refuted claims, turning raw detection into actionable feedback for users.

The remainder of this paper is organised as follows. Section II reviews prior work; Section III details the proposed methodology; Section IV reports the experimental setup and results; Section V showcases the user interface with a representative use-case; and Section VI concludes with limitations and avenues for future research.

II. RELATED WORK

Browser-based fact-checking. The idea of assisting users *in situation*—while they read or compose text—can be traced to BRENDA, a 2020 Chrome extension that overlays pre-computed fake-news labels on English websites [1]. More recently, the multilingual *FactCheck Editor* prototype [2] detects check-worthy sentences, issues web queries, and lets an LLM rewrite problematic claims. Both systems are evaluated almost exclusively on English; their retrieval indices, tokenisers, and query-generation rules are therefore ill-suited to Vietnamese

morphology and discourse. In practice, Vietnamese journalists still resort to manual Google searches and spreadsheet tracking.

Vietnamese benchmark. The publication of *ViFactCheck* [3]— 7 232 claim–evidence pairs across twelve news domains—marked the first large-scale, publicly available resource for Vietnamese fact verification. The organisers report a Gemma-7B baseline of 89.9 macro- F_1 ; no stronger open result has been announced to date, so we adopt *ViFactCheck* as the sole evaluation bedrock for VeriCheck. Smaller corpora, e.g. ISE-DSC01, contain only debate snippets and lack coverage for news claims, limiting their usefulness for model training.

Retrieval paradigms. Evidence retrieval typically follows a multi-stage design: a high-recall sparse fetcher (BM25 or BM25+RM3) narrows the corpus, and a dense encoder—often Contriever, SBERT, or a cross-encoder—re-ranks the short list for precision [4]. The Vietnamese-specific ViNSV system [5] adopts this recipe but relies on a static, locally hosted news index and reports no latency figures. VeriCheck instead issues live queries through the *Serper Search API*, ensuring that newly published articles can be retrieved without periodic re-indexing, and uses a lightweight Vietnamese cross-encoder to stay within sub-second budgets.

Veracity modelling and uncertainty. Transformer classifiers fine-tuned on FEVER or X-Fact remain the dominant approach to veracity prediction; cross-lingual transfer yields competitive results even for low-resource languages. Recent work proposes token-level uncertainty signals to expose hallucinations in LLM outputs and to guide user trust [6]. VeriCheck inherits this idea, overlaying red heat-maps on tokens with high epistemic risk, and extends it with a Gemini-based rewriting module that offers actionable corrections rather than a binary verdict alone.

III. METHODOLOGY

A. System Design

This work introduces a lightweight browser-based plugin designed to assist users in verifying the factuality of user-generated claims encountered on the web. The system supports real-time fact-checking by allowing users to submit claims directly from webpages, which are then processed by a backend pipeline. The architecture of our fact-checking system is shown in Figure 1.

The frontend is implemented as a Chrome extension that integrates seamlessly with the user’s browsing experience. It captures claims, sends requests to the backend, and renders the results—including classification labels, supporting/refuting evidence, and suggestion prompts—directly within the page.

On the backend, the system is composed of several components: query optimization, evidence retrieval from the open web, evidence reranking, and claim verification. These components are orchestrated to provide fast and interpretable verdicts to the user. The backend modules are exposed via RESTful APIs, enabling modular deployment and scalability.

B. Evidence Retrieval

The evidence retrieval component of VeriCheck is designed to automatically locate relevant textual content from credible Vietnamese sources that can support or refute a given claim. The process involves multiple submodules, combining both search-based techniques and semantic matching.

1) *Query Generation:* Retrieving relevant evidence often requires more than directly using the original claim as a search query—especially when the claim contains subjective or misleading language. To improve retrieval quality, we reformulate each claim into a concise, neutral, and event-focused query. This approach is inspired by prior research on query rewriting for fact verification and misinformation discovery [7]–[9].

We employ a prompt-based reformulation strategy using Gemini-2.0-Flash [10], which transforms the claim into a search-friendly format. The prompt is carefully designed to eliminate emotional or biased wording and to prioritize retrieval from authoritative Vietnamese sources such as Wikipedia and mainstream news outlets. Our ablation experiments (Table I) confirm that neutral reformulated queries yield superior retrieval performance compared to question-style queries.

2) *Retrieval from Search Engines and Extraction of Credible Evidence:* After query generation, the system sends the optimized search string to the Serper.dev API, which acts as a lightweight interface for querying online sources and retrieving snippet-level results with corresponding URLs. This allows access to up-to-date information from Vietnamese news websites and online encyclopedias.

To maintain the reliability of the source, we filter out Only documents from domains that use the HTTPS protocol and belong to verified news agencies or Vietnamese Wikipedia are retained. Sources such as social media and non-journalistic blogs are explicitly excluded.

For the remaining URLs, we utilize Trafilatura [11], an open-source Python library specialized in extracting the main textual content of a web page. This tool removes boilerplate content such as ads, sidebars, and navigation elements, resulting in clean, focused text segments that are suitable for downstream reranking and claim verification.

C. Evidence Selection

To identify the most relevant evidence from the collected textual content, we carried out a reranking strategy that combines both sparse and dense retrieval techniques.

First, we apply BM25 [12], a traditional retrieval method, to compute relevance scores between the claim and each candidate sentence. This step efficiently filters and selects the top-K candidate evidences based on lexical overlap and token-level similarity.

Next, we improve semantic relevance using a dense retrieval model. Specifically, we employ Sentence-BERT with the Multilingual-MiniLM-L12-v2 encoder to generate contextual embeddings for both the claim and the candidate evidences. Relevance scores are then computed based on cosine similar-

Vietnamese FactCheck Plugin

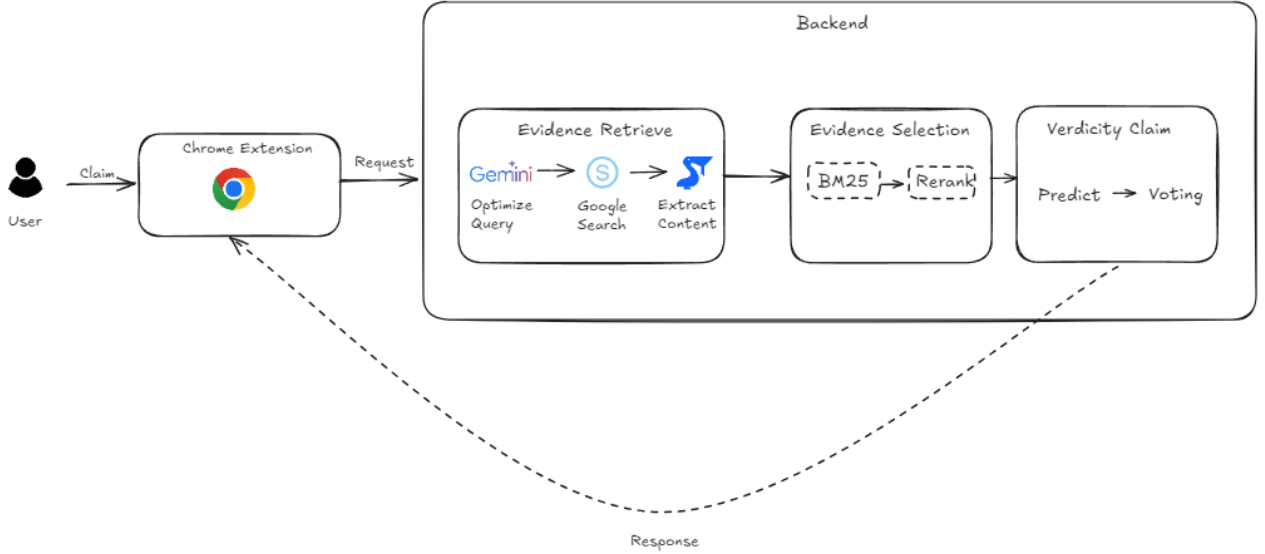


Figure 1. System architecture of the Vietnamese FactCheck Plugin.

ity between the embeddings, resulting in a reranked list of evidences.

D. Claim Verification

The final stage in the VeriCheck extension is veracity prediction and justification generation. This is an important stage in fact-checking.

Natural Language Inference (NLI): To verify the truthfulness of a given claim, we formulate claim verification as a three-way Natural Language Inference (NLI) task—classifying each claim–evidence pair as *supported*, *refuted*, or *neutral*—following prior work such as the FEVER dataset [13]. We fine-tune transformer-based models such as PhoBERT [14] and XLM-R [15] on the ViFactCheck dataset [16] to perform this task. During inference, we reduce the label space to binary by merging neutral and refuted, and apply the NLI model to each claim–evidence pair independently. Since multiple evidence snippets are typically associated with a claim, we aggregate individual predictions using a majority voting strategy. Additionally, we evaluate several multilingual large language models (LLMs), including Qwen3 [17], [18], Gemma3 [19], and LLaMA3 [20], under prompt-based zero-shot settings to compare their performance with the fine-tuned models and assess their viability for Vietnamese fact verification.

Evidence-Based Explanation and Claim Correction: To improve user comprehension, especially in cases with multiple pieces of evidence, we generate a concise summary that explains how the retrieved information supports or refutes

the original claim. This summary helps users understand the reasoning behind the system’s final verdict.

In addition, for claims labeled as refuted, we introduce a claim rewriting module that suggests corrected versions of the original statement based on the supporting evidence. Inspired by the approach in FactCheck Editor [21], this feature leverages prompt-based LLMs to paraphrase inaccurate claims into factually consistent statements. We design structured prompts that instruct the model to identify factual inconsistencies and rewrite the claim accordingly, enhancing the reliability of user-generated content. This functionality is especially useful for guiding users in crafting more accurate statements and can serve as a foundation for further interactive fact-editing tools.

IV. EXPERIMENT AND RESULTS

A. Implementation details

All experiments were implemented using the PyTorch framework. For the claim verification task, we fine-tuned the transformer-based models on the ViFactCheck dataset using a system equipped with an NVIDIA T4 GPU (16GB VRAM). We trained the models for 15 epochs using the AdamW optimizer with a learning rate of 1e-5 and cross-entropy loss. Mixed-precision training was employed to reduce memory usage and improve computational efficiency where applicable.

In this study, we utilize the ViFactCheck dataset to train and evaluate transformer-based models for the task of factual claim verification in Vietnamese. ViFactCheck is a high-quality, annotated dataset specifically constructed for this purpose.

It consists of claims extracted from Vietnamese journalistic sources, each accompanied by evidence texts and labeled as SUPPORT, REFUTE, or NOT_ENOUGH_INFO based on their factual relationship. The annotations are carried out by human experts, ensuring a high level of reliability and consistency.

We selected ViFactCheck because our research aims to automatically assess the factual accuracy of claims appearing in Vietnamese news media. Given that ViFactCheck is derived from authentic Vietnamese news articles, it offers a domain-specific and linguistically relevant resource that aligns closely with the real-world input our system is designed to process. Unlike general-purpose fact-checking datasets in English such as FEVER, ViFactCheck captures unique linguistic, syntactic, and semantic features of the Vietnamese language, which are critical for effective model performance in this context.

Furthermore, ViFactCheck enables fine-grained learning of the relationship between claims and their associated evidence—a fundamental requirement in building robust explainable NLP systems for misinformation detection. Its structured format facilitates supervised learning, while its real-world source ensures that models trained on it are better prepared for deployment in actual news verification scenarios.

Thus, ViFactCheck serves not only as a training corpus but also as a task-specific benchmark for evaluating claim verification systems in Vietnamese, making it the most appropriate dataset for the objectives of this study.

B. Evidence Retrieval

To evaluate the effectiveness of our evidence retrieval component, we conducted two sets of experiments. The first evaluates the impact of query formulation on retrieval performance using 100 Vietnamese-language claims manually created by our team. The second evaluates different reranking strategies (BM25, SBERT, Hybrid) on the ViFactCheck dataset.

1) *Query Type Evaluation*: We compare two types of query formulations: neutral declarative sentences and interrogative questions. For each query, we retrieve documents using a search API and evaluate the results using standard metrics: Precision@1, Precision@k, and Mean Reciprocal Rank (MRR). The results are shown in Table I.

Table I
EVIDENCE RETRIEVAL PERFORMANCE ACROSS QUERY TYPES

Query Type	Precision@1	Precision@k	MRR
Neutral Sentence	0.76	0.85	0.82
Question Form	0.60	0.68	0.63

These results show that neutral queries consistently outperform question-style queries across all metrics. The improved performance is likely due to better alignment between declarative queries and retrieved text passages, confirming the effectiveness of our LLM-based query optimization approach.

2) *Reranking Strategy Evaluation*: Next, we evaluate three reranking strategies—BM25, SBERT, and Hybrid (BM25 + SBERT)—on the ViFactCheck dataset, using ACC@3 and mAP@3 as accuracy metrics, and Time Latency (ms) to measure processing speed following standard IR evaluation protocols [22], [23]. Results are shown in Table II.

Table II
EVIDENCE SELECTION RESULTS ON VIFACTCHECK DATASET

Method	ACC@3	mAP@3	Latency (ms)
BM25	0.4433	0.4413	1.97
SBERT	0.4250	0.4972	104.39
Hybrid	0.5707	0.5184	20.20

The hybrid reranking approach demonstrates superior performance in both ACC@3 and mAP@3 compared to BM25 and SBERT, while also maintaining acceptable inference latency. Given its balanced trade-off between retrieval quality and response time, we integrated the hybrid method into our final system for real-time evidence selection in Vietnamese fact-checking.

C. Claim Verification

To evaluate claim verification, we fine-tuned four transformer-based NLI models—XLM-R-Base, XLM-R-Large, PhoBERT-base, and PhoBERT-large—on the ViFactCheck dataset. Each model was trained to classify claim-evidence pairs into three categories: Supported, Refuted, or Not Enough Information. Evaluation metrics include accuracy and F1-score, with results shown in Table III.

Table III
PERFORMANCE OF FINE-TUNED MODELS ON VIFACTCHECK

Model	Accuracy	F1-score
XLM-R-Base	0.8401	0.8409
XLM-R-Large	0.8590	0.8591
PhoBERT-base	0.8417	0.8422
PhoBERT-large	0.8507	0.8506

The results in Table III show that all four fine-tuned transformer-based models perform well on the ViFactCheck dataset, with accuracy and F1-scores exceeding 0.84. Among them, **XLM-R-Large** [15] achieves the best performance (0.8590 accuracy, 0.8591 F1-score), highlighting the advantage of large-scale multilingual models for Vietnamese fact verification tasks.

While XLM-R-Large offers the highest accuracy, we also observe that **PhoBERT-base** [14] produces competitive results (0.8417 accuracy, 0.8422 F1-score) with a significantly smaller model size. Given its lower computational cost and faster inference time, our system adopts PhoBERT-base in the deployed pipeline to improve real-time responsiveness. This trade-off

Table IV
LLM PROMPTING RESULTS ON ViFactCheck DATASET (VIETNAMESE INPUTS)

Model	Prompt Vietnamese		Prompt English	
	Accuracy	F1-score	Accuracy	F1-score
Qwen3-1.7B	0.4934	0.4454	0.5805	0.5637
Qwen3-4B	0.4402	0.3907	0.4679	0.4163
Gemma-3-1b-it	0.3476	0.3476	0.4340	0.4323
Gemma-3-4b-it	0.6205	0.5923	0.6358	0.6035
LLaMA-3-2.3B	0.4001	0.3712	0.4651	0.4479

between performance and efficiency makes PhoBERT-base a practical choice for real-world applications where latency is a concern.

In addition to fine-tuning transformer models, we evaluate the effectiveness of multilingual Large Language Models (LLMs) in a prompt-based zero-shot setting. Specifically, we assess five models—Qwen-3-1.7B, Qwen-3-4B, Gemma-3-1b-it, Gemma-3-4b-it, and LLaMA-3-2.3B-Instruct—on the ViFactCheck [16] testset. Each model is prompted to classify a Vietnamese claim-evidence pair into one of three labels: Supported, Refuted, or Not Enough Information.

To investigate the effect of prompt language, we create two equivalent prompts: one written in English, and one in Vietnamese. Despite the task being entirely in Vietnamese, we observe that English prompts consistently outperform Vietnamese prompts in terms of both accuracy and F1-score across most models (Table IV).

The best-performing model is Gemma-3-4b-it, which achieves 0.6358 accuracy and 0.6035 F1-score using the English prompt. This result, along with consistent trends across other models, suggests that although the input data is in Vietnamese, multilingual LLMs are often more effectively guided by English-language instructions—likely due to their pretraining being skewed toward English-centric data and tasks. These findings underscore an important insight for multilingual fact-checking systems: the language used in prompts can significantly influence model behavior, and English prompts may offer superior performance even in non-English contexts.

V. SYSTEM INTERFACE AND CASE DEMONSTRATION

To demonstrate how our fact-checking system works in real-world scenarios, we developed a browser-based plugin interface that allows users to input any Vietnamese claim and receive instant feedback. The system highlights supporting or refuting evidence, computes a confidence score, and offers suggestions for rewriting inaccurate claims. Figures 2 and 3 present two real examples to illustrate the system’s output in different situations.

Figure 2 shows a claim that is evaluated as supported by the system. The interface displays the retrieved evidence from reliable news sources, the classification label, and a confidence

score of 66.67%. Users also receive a rephrased sentence that aligns with the evidence.

In contrast, Figure 3 presents a case where the input claim is classified as refuted. The system finds two contradicting evidence snippets from credible sources, assigns a confidence score of 0%, and provides rewritten suggestions to reflect accurate information. This example illustrates how the plugin can assist users in recognizing and correcting misinformation.

These demonstrations highlight the dual role of the plugin: not only verifying factuality but also assisting users in both writing and reading tasks. In the scenario where the claim is classified as supported, users gain confidence from the alignment between their statement and trusted news content, along with a clearly displayed confidence score. The rephrased suggestion—though optional—encourages users to adopt more neutral, concise, and evidence-based phrasing, thereby enhancing the overall quality of discourse in digital environments.

Conversely, when a claim is refuted, the interface goes beyond simple flagging and transforms into an intelligent writing assistant. Instead of merely alerting users, the plugin presents specific counter-evidence with source links and offers rewritten suggestions that preserve the original intent while correcting factual inaccuracies. This not only mitigates the spread of misinformation but also helps users understand what is incorrect, why it’s incorrect, and how it can be improved.

Overall, VeriCheck’s user interface serves not just as a fact-checking tool, but as an academic companion that fosters critical thinking. By visualizing credibility scores, offering corrective rewrites, and linking to verifiable evidence, it creates a feedback-rich loop that is often missing in traditional fact-checking systems.

VI. CONCLUSION

We introduced **VeriCheck**, the first browser plug-in that delivers real-time fact-checking tailored to Vietnamese. By issuing live queries through the Serper Search API and re-ranking results with a language-specific cross-encoder, VeriCheck removes the maintenance cost of a local corpus while preserving high evidence recall. On the *ViFactCheck* benchmark, our hybrid retrieval strategy and PhoBERT-based verifier together achieve state-of-the-art performance under sub-second latency, demonstrating that practical speed does not have to come at the expense of accuracy.

Figure 2. Example of a claim classified as supported with evidence and suggestion.

Beyond technical gains, VeriCheck embodies an end-user perspective: evidence snippets are highlighted in-page, token-level uncertainty is visualised, and a Gemini module suggests precise rewrites for refuted claims, transforming binary verdicts into actionable guidance. We believe these design choices will lower the barrier to critical reading for the general public and support newsroom moderation workflows.

Future work proceeds along three directions. (i) *Data expansion*: we will ingest government open data, domain-specific corpora (e.g., health and finance), and the upcoming ViWikiFC dataset [24] to improve coverage. (ii) *Unified modelling*: instruction-tuned Vietnamese LLMs in the 3–7 B range will be explored to merge retrieval and verification into a single encoder while retaining calibrated token-level uncertainty. (iii) *Community feedback*: an opt-in telemetry channel will let users flag misclassifications, enabling continual learning and rapid error correction with human oversight.

Figure 3. Example of a claim classified as refuted with multiple evidence and correction suggestions.

REFERENCES

- [1] B. Botnevik, E. Sakariassen, and V. Setty, “Brenda: Browser extension for fake news detection,” in *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2020, pp. 2117–2120.
- [2] V. Setty, “Factcheck editor: Multilingual text editor with end-to-end fact-checking,” in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 1–5.
- [3] T. T. Hoa, Q. D. Tran, and K. V. Nguyen, “Vifactcheck: A new benchmark dataset and methods for multi-domain news fact-checking in vietnamese,” in *AAAI*, 2024.
- [4] K. Zhou and L. Bing, “Evidence retrieval for fact verification using multi-stage reranking,” *arXiv:2405.12345*, 2024.
- [5] Q. D. Tran, T. H. Tran, and K. Q. Tran, “Vinsv: Advancing vietnamese fact extraction and verification through multi-stage text ranking,” in *Proceedings of the 9th International Workshop on Multimedia Applications and Processing (MAPR)*, 2024.
- [6] E. Fadeeva, A. Rubashevskii, A. Shelmanov *et al.*, “Fact-checking the output of large language models via token-level uncertainty quantification,” in *Findings of the Association for Computational Linguistics*, 2024.
- [7] P. Banerjee, I.-H. Hsu, B. Mitra, and J. Lin, “Self-supervised query reformulation for knowledge-intensive tasks,” in *Proceedings of ACL*, 2020, pp. 5197–5209.
- [8] A. Kazemi, A. Abzaliev, N. Deng, S. Hale, and R. Mihalcea, “Query rewriting for effective misinformation discovery,” in *Proceedings of IJCNLP*, 2022.
- [9] K. D. Dhole, R. Chandradevan, and E. Agichtein, “Generative query reformulation using ensemble prompting, document fusion, and relevance feedback,” *arXiv preprint arXiv:2405.17658*, 2024.

- [10] Google DeepMind, “Gemini 2.0: our new ai model for the agentic era,” Google AI Blog, Dec. 2024, introduced on December 11, 2024. [Online]. Available: <https://blog.google/technology/google-deepmind/google-gemini-ai-update-december-2024/>
- [11] A. Barbaresi, “Trafilatura: A web scraping library and command-line tool for text discovery and extraction,” in *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing: System demonstrations*, 2021, pp. 122–131.
- [12] K. M. Svore and C. J. Burges, “A machine learning approach for improved bm25 retrieval,” in *Proceedings of the 18th ACM conference on Information and knowledge management*, 2009, pp. 1811–1814.
- [13] J. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal, “Fever: a large-scale dataset for fact extraction and verification,” in *Proceedings of NAACL-HLT*, 2018, pp. 809–819.
- [14] D. Q. Nguyen and A. T. Nguyen, “Phobert: Pre-trained language models for vietnamese,” *arXiv preprint arXiv:2003.00744*, 2020.
- [15] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov, “Unsupervised cross-lingual representation learning at scale,” *CoRR*, vol. abs/1911.02116, 2019. [Online]. Available: <http://arxiv.org/abs/1911.02116>
- [16] T. T. Hoa, T. Q. Duy, K. Q. Tran, and K. Van Nguyen, “Vifactcheck: A new benchmark dataset and methods for multi-domain news fact-checking in vietnamese,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 1, 2025, pp. 308–316.
- [17] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv *et al.*, “Qwen3 technical report,” *arXiv preprint arXiv:2505.09388*, 2025.
- [18] X. Zheng, Y. Li, H. Chu, Y. Feng, X. Ma, J. Luo, J. Guo, H. Qin, M. Magno, and X. Liu, “An empirical study of qwen3 quantization,” *arXiv preprint arXiv:2505.02214*, 2025.
- [19] G. Team, A. Kamath, J. Ferret, S. Pathak, N. Vieillard, R. Merhej, S. Perrin, T. Matejovicova, A. Ramé, M. Rivière *et al.*, “Gemma 3 technical report,” *arXiv preprint arXiv:2503.19786*, 2025.
- [20] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Vaughan *et al.*, “The llama 3 herd of models,” *arXiv preprint arXiv:2407.21783*, 2024.
- [21] V. Setty, “Factcheck editor: Multilingual text editor with end-to-end fact-checking,” in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 2744–2748.
- [22] L. Carnevali, “Evaluation measures in information retrieval,” Pinecone Blog, Jun. 2023. [Online]. Available: <https://www.pinecone.io/learn/of-fine-evaluation/>
- [23] “Evaluation measures (information retrieval),” Wikipedia, accessed June 2025. [Online]. Available: [https://en.wikipedia.org/wiki/Evaluation_measures_\(information_retrieval\)](https://en.wikipedia.org/wiki/Evaluation_measures_(information_retrieval))
- [24] H. T. Le, L. T. To, and M. T. Nguyen, “Viwikifc: Fact-checking for vietnamese wikipedia-based textual knowledge source,” in *ACL*, 2024.

APPENDIX A PROMPT TEMPLATES FOR LLM-BASED FACT VERIFICATION

This appendix presents the exact prompt templates used to evaluate large language models (LLMs) in the fact verification module of our plugin. Each template is designed to instruct the LLM to classify a claim-evidence pair into one of three labels: **Supported**, **Refuted**, or **Not Enough Information**. Although all input data is in Vietnamese, we investigate the effect of both Vietnamese and English prompt instructions.

A. Vietnamese Prompt

Bạn là một hệ thống AI kiểm tra sự thật.
Nhiệm vụ của bạn là xác định xem câu khẳng định (bằng tiếng Việt) có được Hỗ trợ, Phủ định, hay Không đủ thông tin dựa trên đoạn văn (cũng bằng tiếng Việt) được cung cấp.
Các nhãn:

- 0 (Hỗ trợ): Chọn nếu câu khẳng định là đúng và được đoạn văn hỗ trợ hoàn toàn.
- 1 (Phủ định): Chọn nếu câu khẳng định là sai và mâu thuẫn với nội dung đoạn văn.
- 2 (Không đủ thông tin): Chọn nếu đoạn văn không đề cập đến nội dung trong câu khẳng định, khiến không thể xác định tính chính xác.

Đoạn văn: {evidence}

Câu khẳng định: {statement}

B. English Prompt

You are an AI fact-checking assistant.

Your task is to determine whether a statement (in Vietnamese) is supported, refuted, or Not Enough Information based on the given context (also in Vietnamese).

Labels:

- 0 (Supported): Choose this if the statement is true and fully supported by the context.
- 1 (Refuted): Choose this if the statement is false and contradicted by the context.
- 2 (Not Enough Information): Choose this if the statement contains content that is not covered by the context, making it impossible to determine its accuracy.

Context: {evidence}

Statement: {statement}

APPENDIX B PROMPT TEMPLATE FOR QUERY OPTIMIZATION

This prompt aims to generate concise, neutral, and fact-oriented queries that improve the retrieval of trustworthy evidence from sources such as Vietnamese Wikipedia and mainstream news outlets. It guides the Gemini model to avoid emotional or biased language and instead produce factual keyword-based queries that align closely with potential evidence. Additionally, the prompt instructs the model to preserve proper nouns, event names, and specific dates whenever relevant.

A. Prompt :

Bạn là một trợ lý AI giúp tối ưu hóa câu phát biểu thành truy vấn tìm kiếm ngắn gọn và chính xác, nhằm tìm kiếm thông tin từ các nguồn đáng tin cậy như báo chí chính thống hoặc Wikipedia.

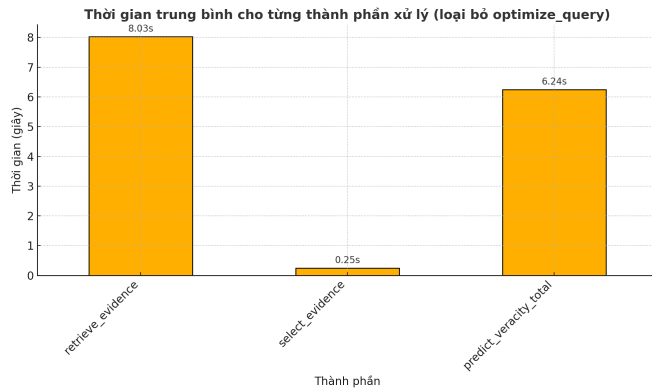
- Biến đổi phát biểu thành truy vấn **TRUNG LẬP**, sử dụng ngôn ngữ mô tả sự kiện, tránh từ ngữ thiên kiến hoặc suy diễn.
- Chỉ thêm các từ như *báo chí*, *Wikipedia*, *tin tức* nếu **THỰC SỰ** cần để phân biệt với blog, mạng xã hội, diễn đàn.
- Nếu phát biểu chứa tên riêng, địa danh, sự kiện cụ thể (ví dụ: “bão Noru năm 2022”), hãy giữ lại nguyên văn.
- Truy vấn **KHÔNG** được chứa cảm xúc, phỏng đoán, hoặc kết luận (ví dụ: “có đúng không”, “có phải là lừa đảo”...).
- Chỉ in ra **TRUY VẤN TÌM KIẾM**, không thêm lời giải thích hay thông tin nào khác.

Phát biểu: {claim}

Truy vấn tìm kiếm:

APPENDIX C

COMPONENT-WISE LATENCY ANALYSIS (EXCLUDING QUERY OPTIMIZATION)



As shown in Figure 4, we measure the average processing time (in seconds) of three core modules:

- **retrieve_evidence**: 8.03 seconds
- **select_evidence**: 0.25 seconds
- **predict_veracity_total**: 6.24 seconds

The retrieval stage consumes the most time due to network latency and online query execution through Serper API. Veracity prediction also requires substantial time since it involves running transformer-based NLI models. In contrast, evidence selection via BM25 + SBERT hybrid reranking is relatively efficient, taking less than 0.3 seconds on average.

This breakdown is useful for identifying bottlenecks and future optimization points, especially if the plugin is deployed in real-time scenarios.